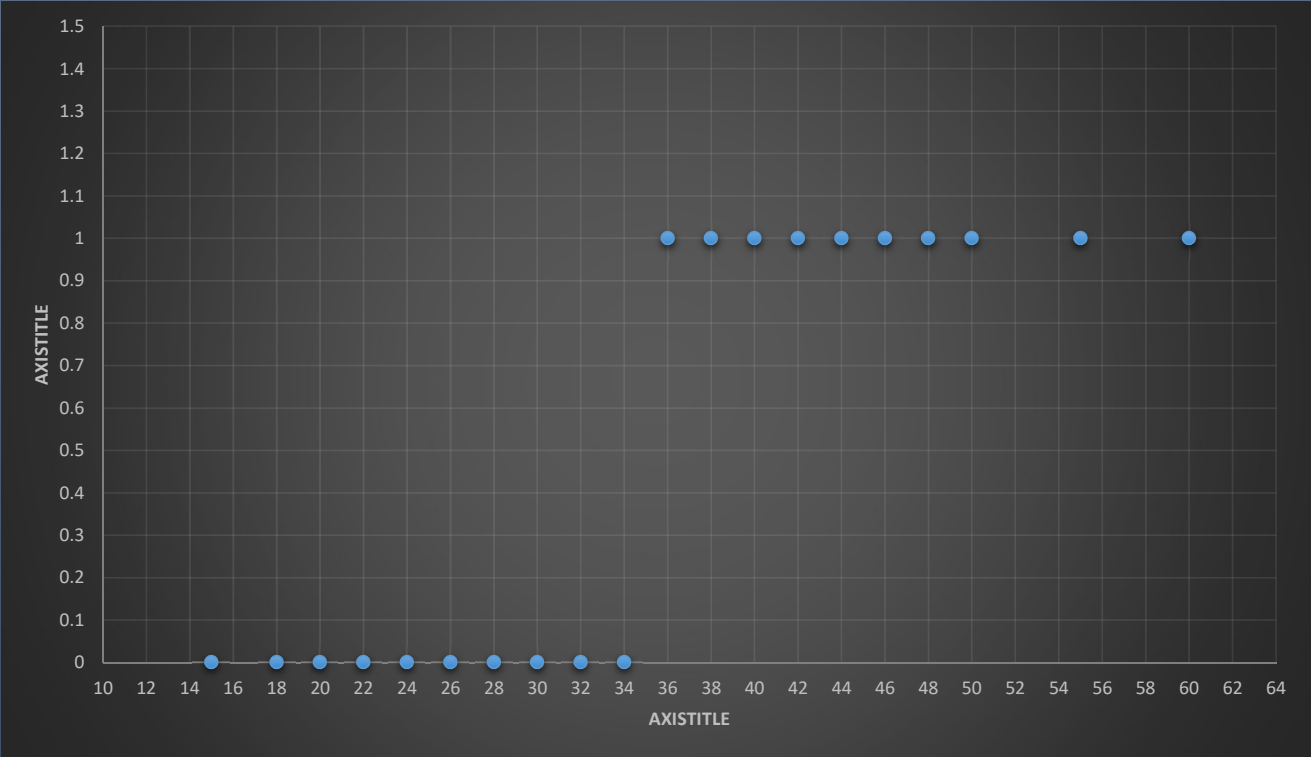


Monthly Income (₹1000s)	Actual Loan Approved	$mx+c$	z	AfterSig moid	0and1	Final Correct
15	0					
18	0					
20	0					
22	0					
24	0					
26	0					
28	0					
30	0					
32	0					
34	0					
36	1					
38	1					
40	1					
42	1					
44	1					
46	1					
48	1					
50	1					
55	1					
60	1					

MonthlyIncome (₹1000s)	LoanApproved
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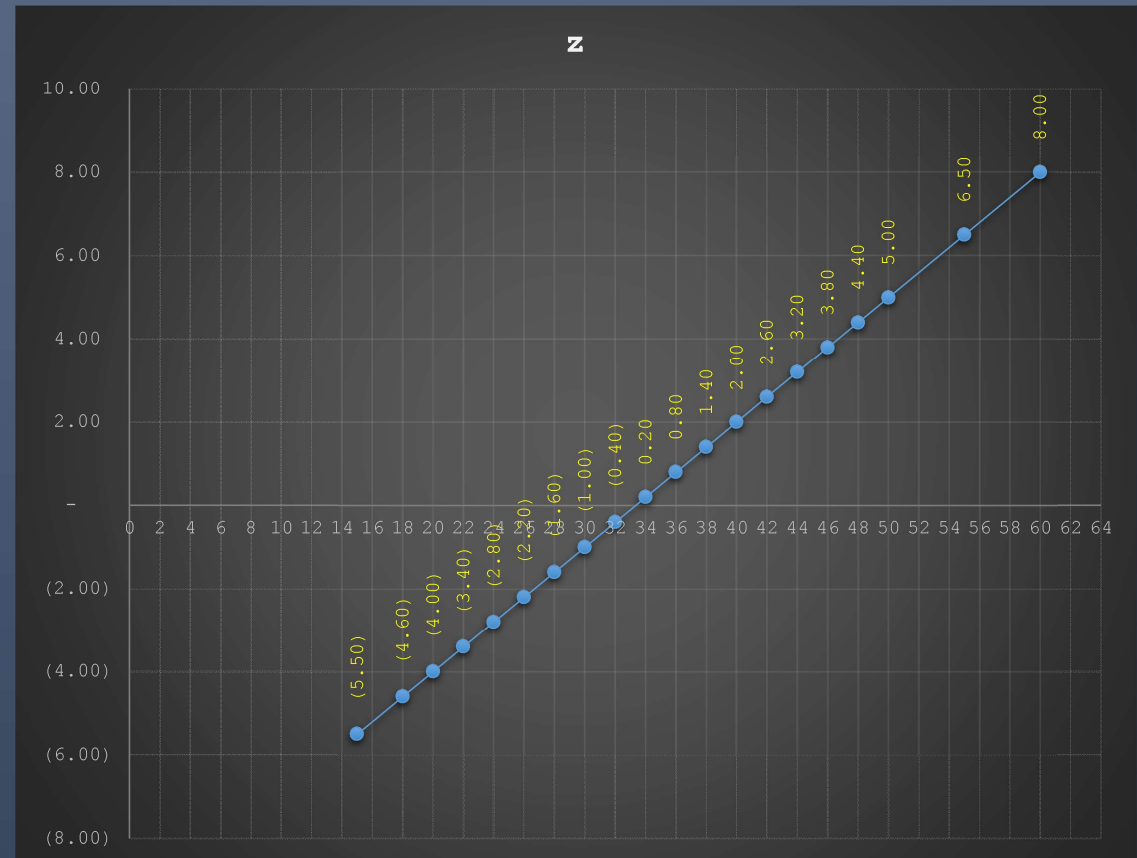
15	0
18	0
20	0
22	0
24	0
26	0
28	0
30	0
32	0
34	0
36	1
38	1
40	1
42	1
44	1
46	1
48	1
50	1
55	1
60	1



Intercept (c)= -10 Slope = 0.3

Monthly Income (₹1000s)	Loan Approved	$mx+c$	z
15	0	$= (0.3 \times 15) + -10$	-5.50
18	0	$= (0.3 \times 18) + -10$	-4.60
20	0	$= (0.3 \times 20) + -10$	-4.00
22	0	$= (0.3 \times 22) + -10$	-3.40
24	0	$= (0.3 \times 24) + -10$	-2.80
26	0	$= (0.3 \times 26) + -10$	-2.20
28	0	$= (0.3 \times 28) + -10$	-1.60
30	0	$= (0.3 \times 30) + -10$	-1.00
32	0	$= (0.3 \times 32) + -10$	-0.40
34	0	$= (0.3 \times 34) + -10$	0.20
36	1	$= (0.3 \times 36) + -10$	0.80
38	1	$= (0.3 \times 38) + -10$	1.40
40	1	$= (0.3 \times 40) + -10$	2.00
42	1	$= (0.3 \times 42) + -10$	2.60
44	1	$= (0.3 \times 44) + -10$	3.20
46	1	$= (0.3 \times 46) + -10$	3.80
48	1	$= (0.3 \times 48) + -10$	4.40
50	1	$= (0.3 \times 50) + -10$	5.00
55	1	$= (0.3 \times 55) + -10$	6.50
60	1	$= (0.3 \times 60) + -10$	8.00

$$z = mx + c$$



Monthly Income (₹1000s)	Loan Approved	mx+c	z	After Sigmoid
15	0	=(0.3*15)+-10	-5.50	0.00
18	0	=(0.3*18)+-10	-4.60	0.01
20	0	=(0.3*20)+-10	-4.00	0.02
22	0	=(0.3*22)+-10	-3.40	0.03
24	0	=(0.3*24)+-10	-2.80	0.06
26	0	=(0.3*26)+-10	-2.20	0.10
28	0	=(0.3*28)+-10	-1.60	0.17
30	0	=(0.3*30)+-10	-1.00	0.27
32	0	=(0.3*32)+-10	-0.40	0.40
34	0	=(0.3*34)+-10	0.20	0.55
36	1	=(0.3*36)+-10	0.80	0.69
38	1	=(0.3*38)+-10	1.40	0.80
40	1	=(0.3*40)+-10	2.00	0.88
42	1	=(0.3*42)+-10	2.60	0.93
44	1	=(0.3*44)+-10	3.20	0.96
46	1	=(0.3*46)+-10	3.80	0.98
48	1	=(0.3*48)+-10	4.40	0.99
50	1	=(0.3*50)+-10	5.00	0.99
55	1	=(0.3*55)+-10	6.50	1.00
60	1	=(0.3*60)+-10	8.00	1.00

$$\text{SigmoidActivation} = \frac{1}{1 + e^{-z}}$$

$$e = 2.718$$

$$= \frac{1}{1 + 2.718^{-(-5.50)}}$$

$$= 0.00$$

$$= \frac{1}{1 + 2.718^{-(-4.60)}}$$

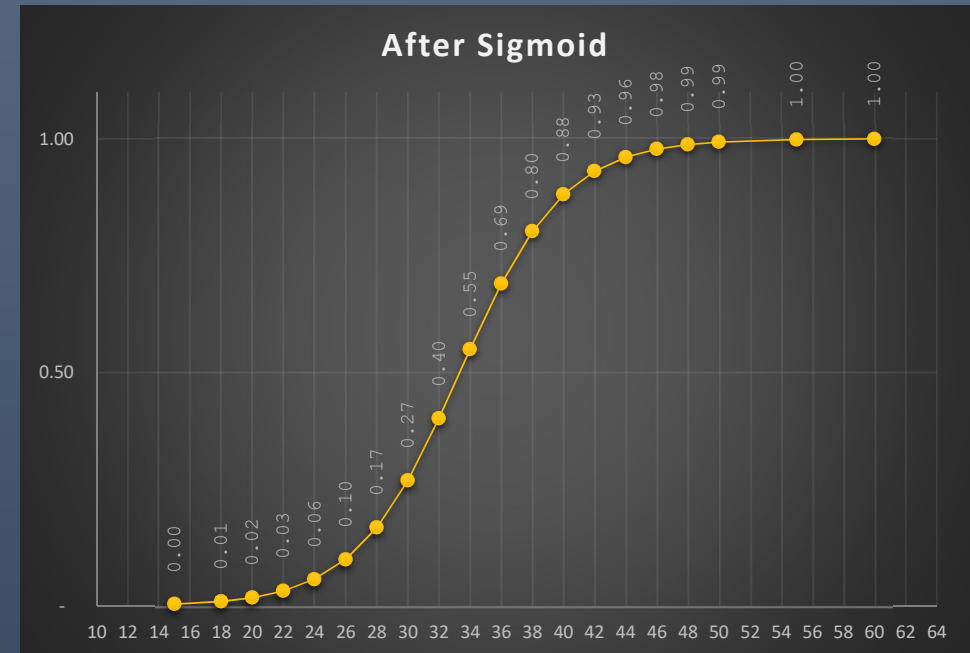
$$= 0.01$$

$$= \frac{1}{1 + 2.718^{-(-4.00)}}$$

$$= 0.02$$

Intercept (c)= -10 Slope = 0.3

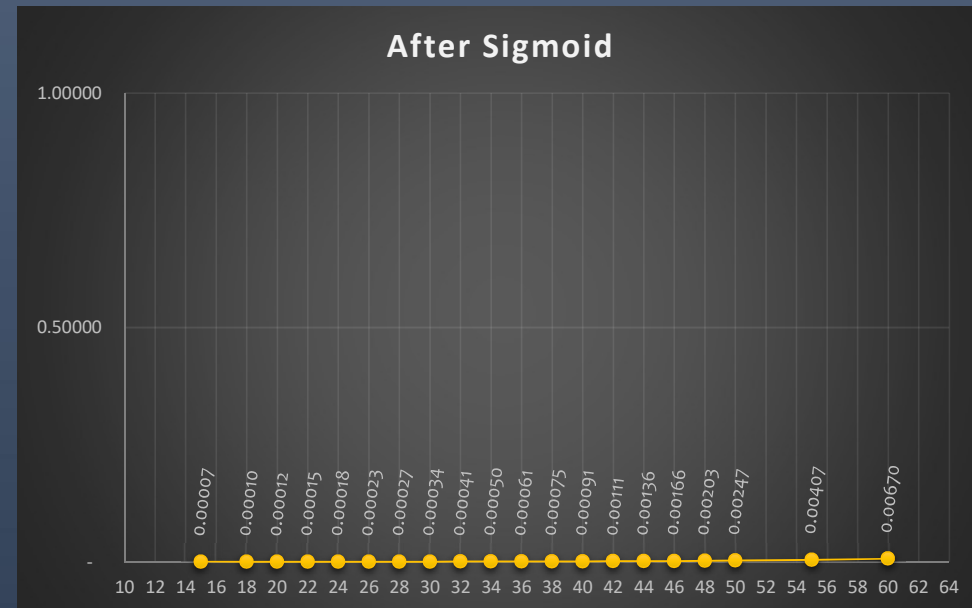
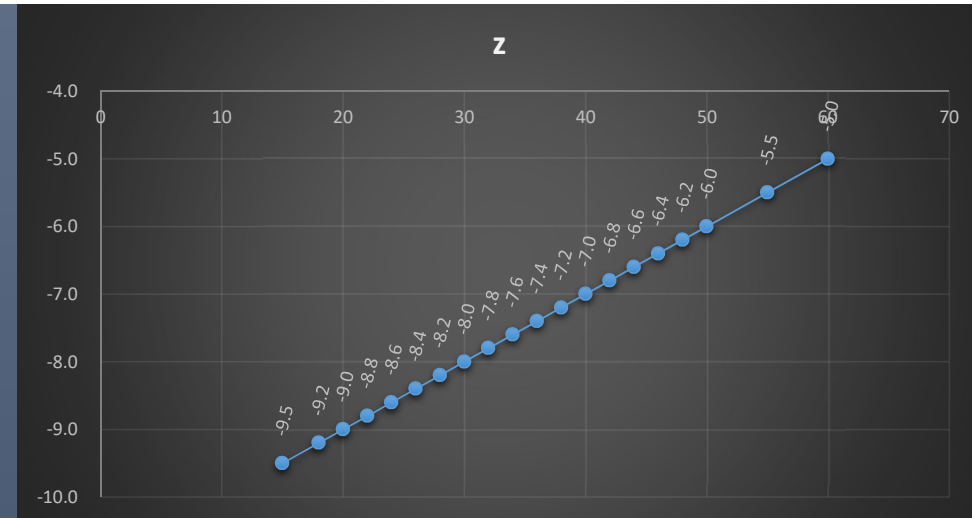
Monthly Income (₹1000s)	Loan Approved	mx+C	z	After Sigmoid
15	0	= (0.3*15)+-10	-5.50	0.00
18	0	= (0.3*18)+-10	-4.60	0.01
20	0	= (0.3*20)+-10	-4.00	0.02
22	0	= (0.3*22)+-10	-3.40	0.03
24	0	= (0.3*24)+-10	-2.80	0.06
26	0	= (0.3*26)+-10	-2.20	0.10
28	0	= (0.3*28)+-10	-1.60	0.17
30	0	= (0.3*30)+-10	-1.00	0.27
32	0	= (0.3*32)+-10	-0.40	0.40
34	0	= (0.3*34)+-10	0.20	0.55
36	1	= (0.3*36)+-10	0.80	0.69
38	1	= (0.3*38)+-10	1.40	0.80
40	1	= (0.3*40)+-10	2.00	0.88
42	1	= (0.3*42)+-10	2.60	0.93
44	1	= (0.3*44)+-10	3.20	0.96
46	1	= (0.3*46)+-10	3.80	0.98
48	1	= (0.3*48)+-10	4.40	0.99
50	1	= (0.3*50)+-10	5.00	0.99
55	1	= (0.3*55)+-10	6.50	1.00
60	1	= (0.3*60)+-10	8.00	1.00



Monthly Income (₹1000s)	Actual Loan Approved	$mx+c$	$mx+c$	After Sigmoid	0 and 1	Final Correct
15	0	$= (0.3 * 15) + -10$	-5.50	0.00	0	Correct
18	0	$= (0.3 * 18) + -10$	-4.60	0.01	0	Correct
20	0	$= (0.3 * 20) + -10$	-4.00	0.02	0	Correct
22	0	$= (0.3 * 22) + -10$	-3.40	0.03	0	Correct
24	0	$= (0.3 * 24) + -10$	-2.80	0.06	0	Correct
26	0	$= (0.3 * 26) + -10$	-2.20	0.10	0	Correct
28	0	$= (0.3 * 28) + -10$	-1.60	0.17	0	Correct
30	0	$= (0.3 * 30) + -10$	-1.00	0.27	0	Correct
32	0	$= (0.3 * 32) + -10$	-0.40	0.40	0	Correct
34	0	$= (0.3 * 34) + -10$	0.20	0.55	0	Correct
36	1	$= (0.3 * 36) + -10$	0.80	0.69	0	-
38	1	$= (0.3 * 38) + -10$	1.40	0.80	1	Correct
40	1	$= (0.3 * 40) + -10$	2.00	0.88	1	Correct
42	1	$= (0.3 * 42) + -10$	2.60	0.93	1	Correct
44	1	$= (0.3 * 44) + -10$	3.20	0.96	1	Correct
46	1	$= (0.3 * 46) + -10$	3.80	0.98	1	Correct
48	1	$= (0.3 * 48) + -10$	4.40	0.99	1	Correct
50	1	$= (0.3 * 50) + -10$	5.00	0.99	1	Correct
55	1	$= (0.3 * 55) + -10$	6.50	1.00	1	Correct
60	1	$= (0.3 * 60) + -10$	8.00	1.00	1	Correct

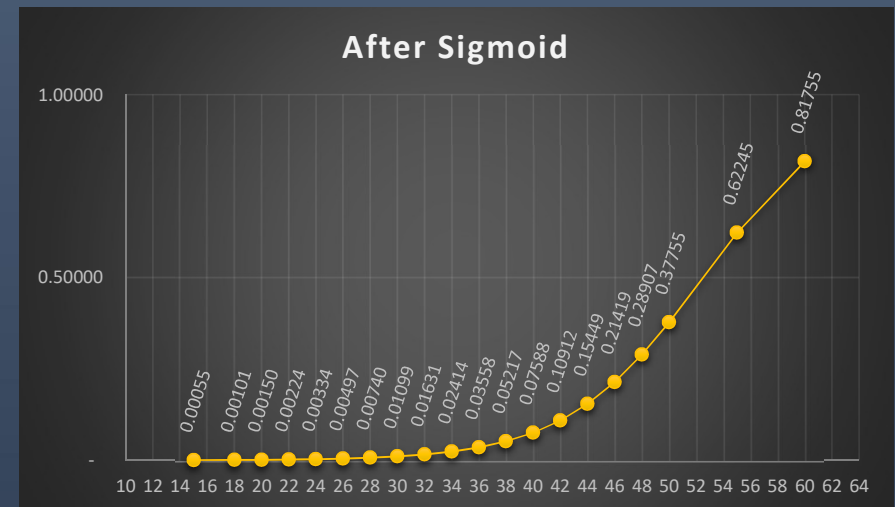
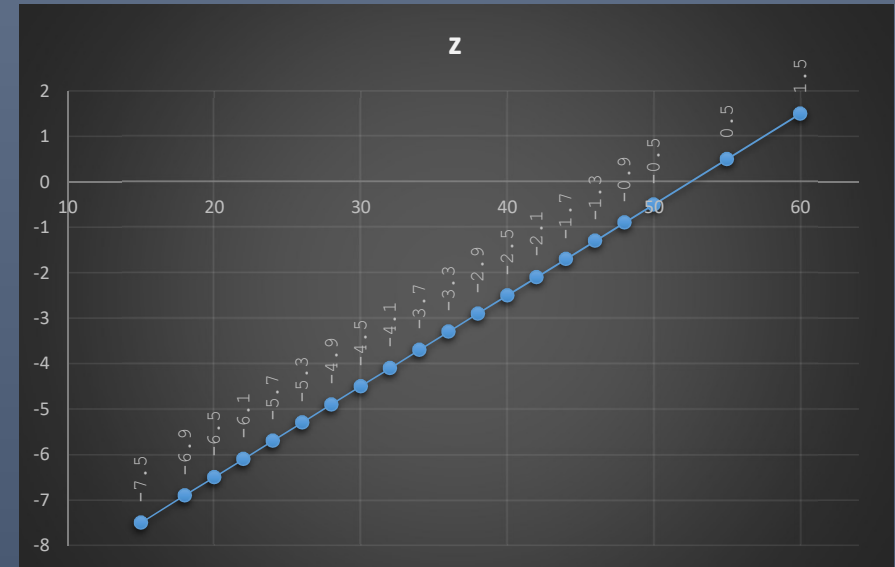
Intercept (c) = -11 Slope = 0.2

Monthly Income (₹1000s)	Actual Loan Approved	z	After Sigmoid	0and1	Final Correct
15	0	-9.5	0.00007	0	Correct
18	0	-9.2	0.00010	0	Correct
20	0	-9.0	0.00012	0	Correct
22	0	-8.8	0.00015	0	Correct
24	0	-8.6	0.00018	0	Correct
26	0	-8.4	0.00023	0	Correct
28	0	-8.2	0.00027	0	Correct
30	0	-8.0	0.00034	0	Correct
32	0	-7.8	0.00041	0	Correct
34	0	-7.6	0.00050	0	Correct
36	1	-7.4	0.00061	0	-
38	1	-7.2	0.00075	0	-
40	1	-7.0	0.00091	0	-
42	1	-6.8	0.00111	0	-
44	1	-6.6	0.00136	0	-
46	1	-6.4	0.00166	0	-
48	1	-6.2	0.00203	0	-
50	1	-6.0	0.00247	0	-
55	1	-5.5	0.00407	0	-
60	1	-5.0	0.00670	0	-



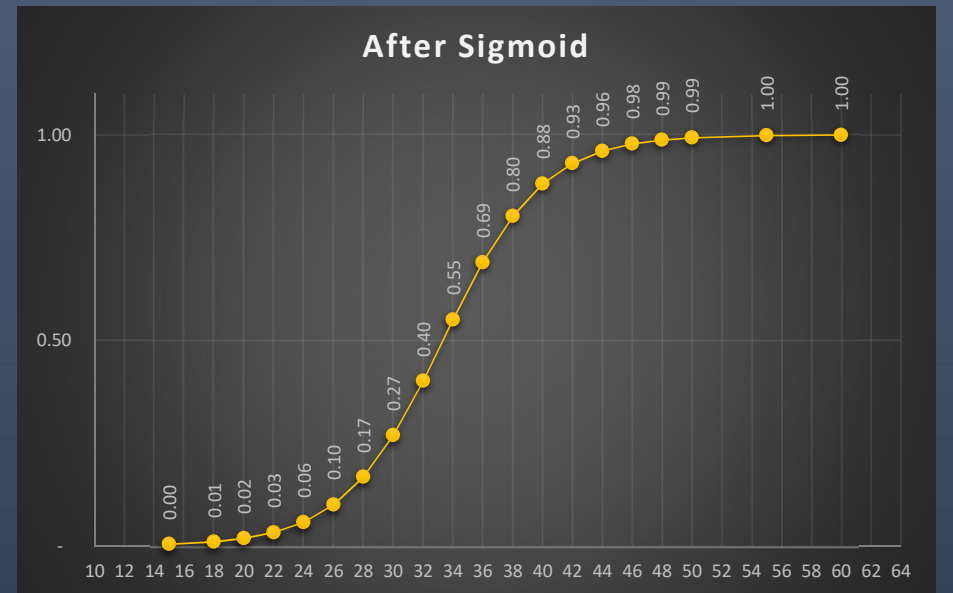
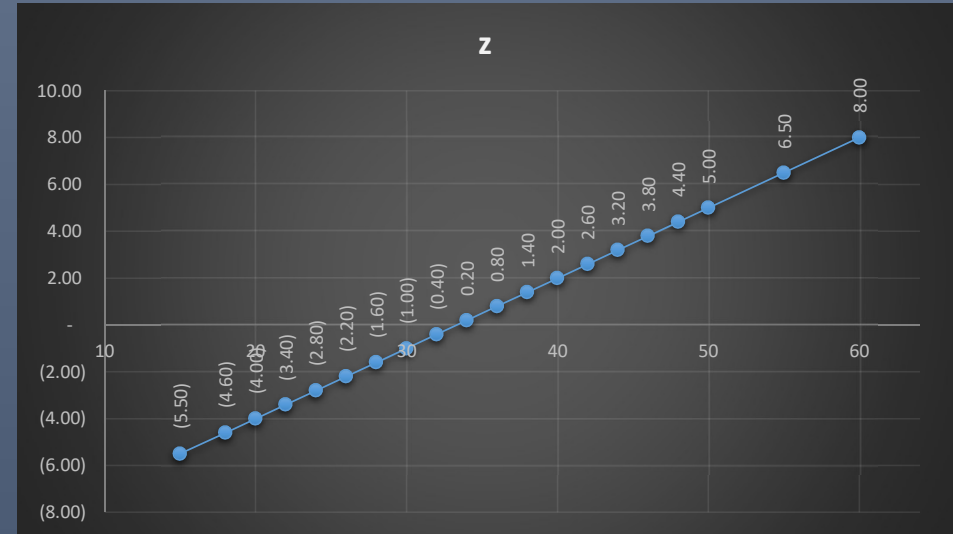
Intercept (c) = -10.5 Slope = 0.2

Monthly Income (₹1000s)	Actual Loan Approved	z	After Sigmoid	0and1	Final Correct
15	0	-7.5	0.00055	0	Correct
18	0	-6.9	0.00101	0	Correct
20	0	-6.5	0.00150	0	Correct
22	0	-6.1	0.00224	0	Correct
24	0	-5.7	0.00334	0	Correct
26	0	-5.3	0.00497	0	Correct
28	0	-4.9	0.00740	0	Correct
30	0	-4.5	0.01099	0	Correct
32	0	-4.1	0.01631	0	Correct
34	0	-3.7	0.02414	0	Correct
36	1	-3.3	0.03558	0	-
38	1	-2.9	0.05217	0	-
40	1	-2.5	0.07588	0	-
42	1	-2.1	0.10912	0	-
44	1	-1.7	0.15449	0	-
46	1	-1.3	0.21419	0	-
48	1	-0.9	0.28907	0	-
50	1	-0.5	0.37755	0	-
55	1	0.5	0.62245	1	Correct
60	1	1.5	0.81755	1	Correct



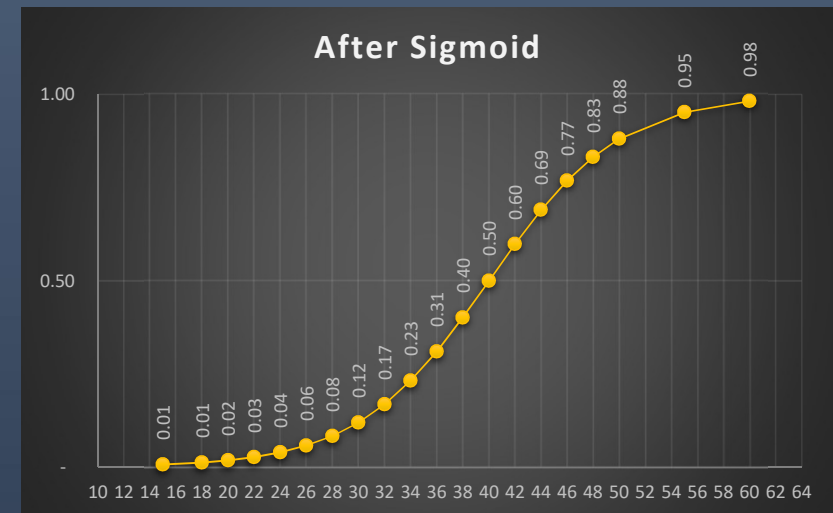
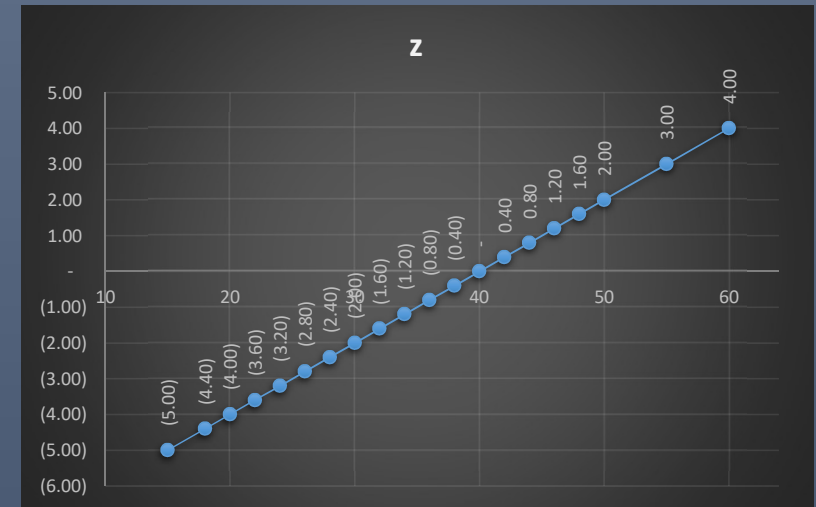
Intercept (c) = -10 Slope = 0.3

Monthly Income (₹1000s)	Actual Loan Approved	z	After Sigmoid	0and1	Final Correct
15	0	-5.50	0.00	0	Correct
18	0	-4.60	0.01	0	Correct
20	0	-4.00	0.02	0	Correct
22	0	-3.40	0.03	0	Correct
24	0	-2.80	0.06	0	Correct
26	0	-2.20	0.10	0	Correct
28	0	-1.60	0.17	0	Correct
30	0	-1.00	0.27	0	Correct
32	0	-0.40	0.40	0	Correct
34	0	0.20	0.55	1	-
36	1	0.80	0.69	1	Correct
38	1	1.40	0.80	1	Correct
40	1	2.00	0.88	1	Correct
42	1	2.60	0.93	1	Correct
44	1	3.20	0.96	1	Correct
46	1	3.80	0.98	1	Correct
48	1	4.40	0.99	1	Correct
50	1	5.00	0.99	1	Correct
55	1	6.50	1.00	1	Correct
60	1	8.00	1.00	1	Correct



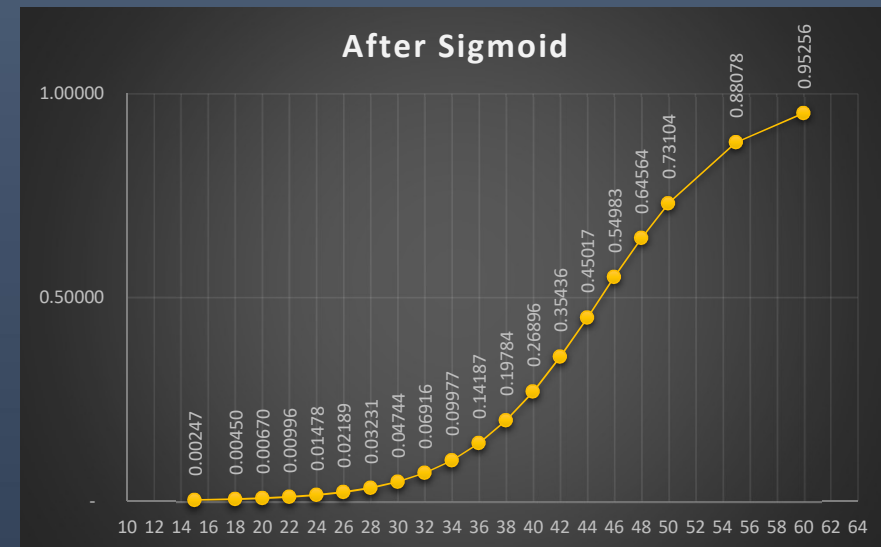
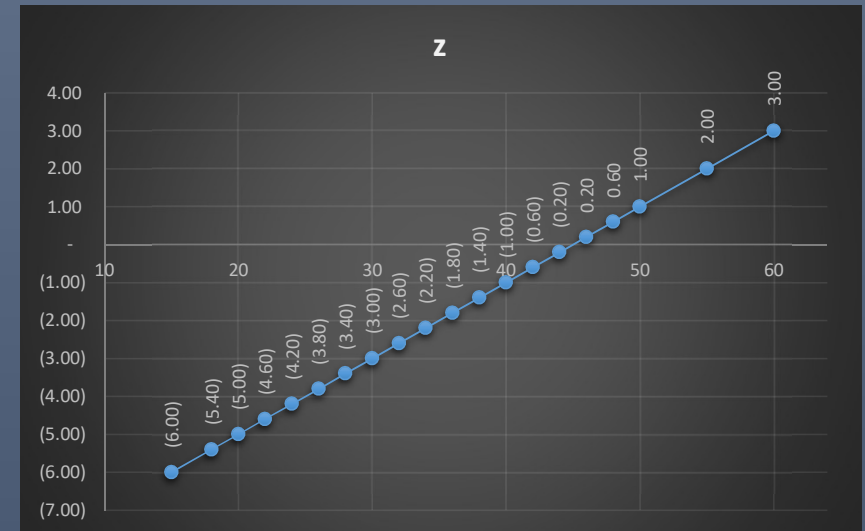
Intercept (c) = -8 Slope = 0.2

Monthly Income (₹1000s)	Actual Loan Approved	z	After Sigmoid	0and1	Final Correct
15	0	-5.00	0.01	0	Correct
18	0	-4.40	0.01	0	Correct
20	0	-4.00	0.02	0	Correct
22	0	-3.60	0.03	0	Correct
24	0	-3.20	0.04	0	Correct
26	0	-2.80	0.06	0	Correct
28	0	-2.40	0.08	0	Correct
30	0	-2.00	0.12	0	Correct
32	0	-1.60	0.17	0	Correct
34	0	-1.20	0.23	0	Correct
36	1	-0.80	0.31	0	-
38	1	-0.40	0.40	0	-
40	1	-	0.50	1	Correct
42	1	0.40	0.60	1	Correct
44	1	0.80	0.69	1	Correct
46	1	1.20	0.77	1	Correct
48	1	1.60	0.83	1	Correct
50	1	2.00	0.88	1	Correct
55	1	3.00	0.95	1	Correct
60	1	4.00	0.98	1	Correct

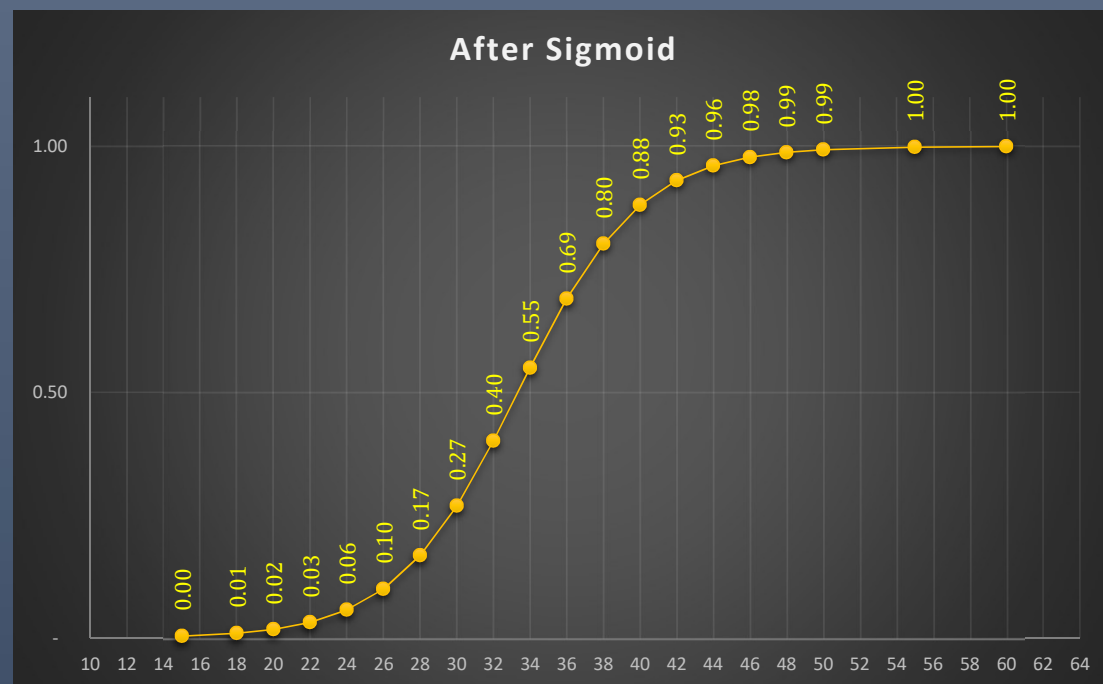


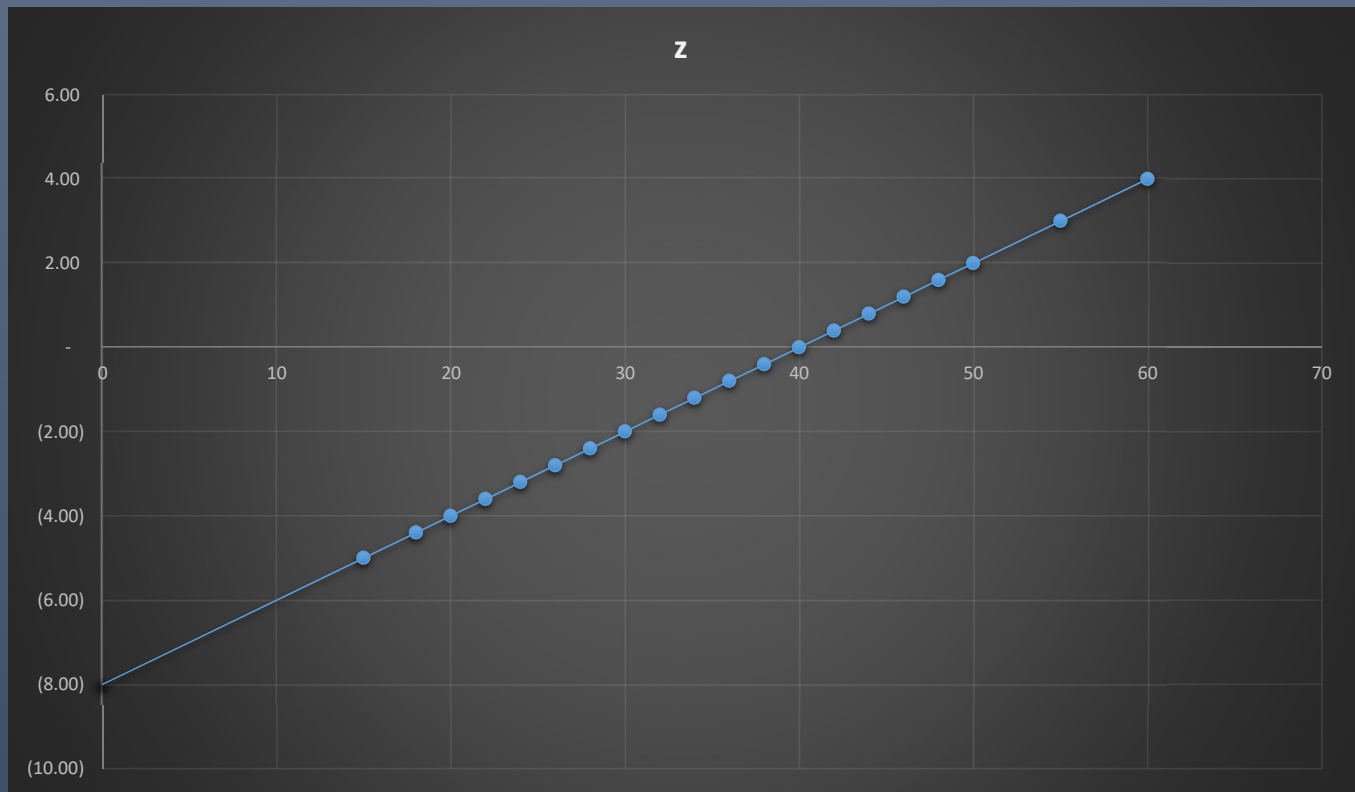
Intercept (c) = -9 Slope = 0.2

Monthly Income (₹1000s)	Actual Loan Approved	z	After Sigmoid	0 and 1	Final Correct
15	0	-6.00	0.00247	0	Correct
18	0	-5.40	0.00450	0	Correct
20	0	-5.00	0.00670	0	Correct
22	0	-4.60	0.00996	0	Correct
24	0	-4.20	0.01478	0	Correct
26	0	-3.80	0.02189	0	Correct
28	0	-3.40	0.03231	0	Correct
30	0	-3.00	0.04744	0	Correct
32	0	-2.60	0.06916	0	Correct
34	0	-2.20	0.09977	0	Correct
36	1	-1.80	0.14187	0	-
38	1	-1.40	0.19784	0	-
40	1	-1.00	0.26896	0	-
42	1	-0.60	0.35436	0	-
44	1	-0.20	0.45017	0	-
46	1	0.20	0.54983	1	Correct
48	1	0.60	0.64564	1	Correct
50	1	1.00	0.73104	1	Correct
55	1	2.00	0.88078	1	Correct
60	1	3.00	0.95256	1	Correct



Intercept	slope	Correct	Incorrect
-11.00	0.10	10	10
-10.50	0.20	12	8
-10.00	0.30	19	1
-8.00	0.20	18	2
-9.00	0.20	15	5





z

